

volutional topology: the input low-resolution image is fed to 5×5 convolutional layer performing feature extraction, followed by five 3×3 convolutions, one 3×3 convolutional and one pixel-shuffle layer reshaping the output to the target resolution (Fig. 7, right image). No residual blocks, skip connections, up-sampling or even addition operations are used in the model to improve its speed. During the training stage, each single convolutional layer is decoupled into a group of three operations:

$$y = x + \text{conv}_{1 \times 1}(x) + \text{conv}_{3 \times 3}(x),$$

followed by *ReLU* activations (Fig. 7, left image). After the end of the training, the parameters of each convolutional group are folded to get a single convolution [10].

The network was trained with a batch size of 32 on 64×64 pixel input images. L_1 loss was used as a target metric, model parameters were optimized for 600 epochs using Adam with an initial learning rate of $5e-4$ decreases by half every 200 epochs. Model quantization was performed with TensorFlow's post-training quantization tools, clipped *ReLU* was used after the last convolutional layer to avoid incorrect outputs normalization.

4.6. mju_gogogo

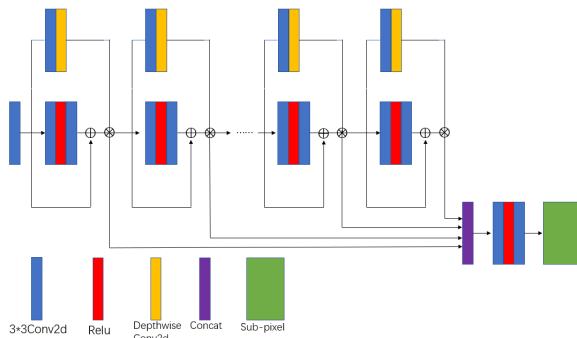


Figure 8. EDSR-based model with attention blocks proposed by mju_gogogo team.

The model architecture proposed by mju_gogogo team is presented in Fig. 8. The model is based on the EDSR [41] design with additional spatial attention blocks. The network is relatively large — it consists of six residual blocks, each one multiplied by the outputs of the corresponding attention unit, one 1×1 convolutional layer reducing the number of filters from 384 to 64, one global skip connection, three 3×3 convolutions and one *depth-to-space* layer. The model was first trained to minimize the *mean absolute error* (MAE) with Adam optimizer for 750 epochs, and then fine-tuned using quantized-aware training and the *MSE* loss for another 75 epochs.

5. Additional Literature

An overview of the past challenges on mobile-related tasks together with the proposed solutions can be found in the following papers:

- Image Super-Resolution: [31, 44, 6, 52]
- Learned End-to-End ISP: [28, 32]
- Perceptual Image Enhancement: [31, 26]
- Bokeh Effect Rendering: [24, 30]
- Image Denoising: [1, 2]

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